
REPORT

Optimisation-Induced Divergence in GCC Security Plugins

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KEY FINDING: High-level optimisations (-O2, -O3, LTO, VECT) consistently erase or inline critical security gates across all six tested GCC plugins, creating a risk of silently omitted protections in production builds.

1. Overview

This report summarises findings from a differential testing campaign targeting six security-critical GCC plugins in the Linux Kernel Hardening suite. Unoptimized (BASE) builds were compared against optimisation variants (-O2, -O3, LTO, VECT) to identify structural divergence that may threaten the integrity of security instrumentation.

While no functional crashes were observed, aggressive optimisation passes were found to erase or inline critical security gates. This creates scenarios where protections such as stack wiping or structure randomisation may be silently omitted in production builds without triggering any visible failures.

Scope of testing:

- Plugins tested: arm_ssp, latent_entropy, randomize_layout, sancov, stack_leak, struct_leak
- Optimization variants: -O0 (BASE), -O2, -O3, LTO, VECT
- Test corpus: yarpgen-generated reproducer files
- Compiler: GCC (versions compatible with provided plugin headers)

2. Comparative Risk Matrix

The table below provides a high-level view of how each plugin responded to aggressive optimization passes. All six plugins received a Likely Bug verdict based on the combined impact of instrumentation erasure and branch elimination.

Plugin	Symbols Erased	Branch Eliminations	Risk Verdict	Primary Impact
arm_ssp	32	3	Likely Bug	Stack protection logic bypass
latent_entropy	21	73	Likely Bug	Entropy source stripping
randomize_layout	13	73	Likely Bug	Deterministic memory layouts
sancov	9	24	Likely Bug	Coverage blind spots
stack_leak	20	105	Likely Bug	Failure to wipe leaky stack frames
struct_leak	12	43	Likely Bug	Uninitialized memory exposure

Note: 'Symbols Erased' refers to security-critical functions that disappeared from optimised binaries. 'Branch Eliminations' refers to conditional checks resolved as constants by the optimiser.

3. Detailed Technical Findings

3.1 Instrumentation Erasure - Disappeared Symbols

Under high-optimisation variants, the following core functions which contain the actual security logic disappeared from the compiled binary. While inlining is a contributing factor, the concern is that Inter-Procedural Analysis (IPA) may be merging these functions into a state where the protection is effectively a no-op.

Affected plugins and erased functions:

- `stack_leak`: `large_stack_frame()`, `is_alloca()`, `stackleak_cleanup_execute()`
- `latent_entropy`: `latent_entropy_execute()`, `create_latent_entropy_decl()`
- `struct_leak`: `initialize()`, `structleak_execute()`
- `randomize_layout`: `relayout_struct()`, `randomize_type()`

RISK: If security functions are silently inlined and subsequently removed by the optimizer (e.g., via Dead Store Elimination), the protection is absent without any compile-time warning.

3.2 Static Gate Resolution - Branch Elimination

The optimizer resolved security branch conditions as compile-time constants in over 340 instances across the plugin suite. This means the compiler 'proved' under its optimisation model that certain code paths would never be taken.

Critical case: `stack_leak`

105 branches were eliminated in the `stack_leak` plugin alone. If the compiler determines that a branch leading to a stack-wipe is unreachable based on optimisation assumptions rather than runtime behavior the security invariant is permanently broken for that compilation unit.

3.3 Inlining of Opaque Logic

In `struct_leak`, the `initialize()` function responsible for zeroing sensitive data structures was flagged for inlining of complex logic. Once inlined, it becomes subject to Dead Store Elimination (DSE), which may remove the zeroing operations as 'useless writes' if the compiler cannot observe any subsequent reads of the data.

CONSEQUENCE: Traditional auditing cannot easily verify whether the zeroing actually takes place in the optimized binary. This creates a 'black box' where sensitive memory may remain unwiped.

4. Reproducibility & Environment

4.1 Toolchain Configuration

Compiler: GCC (versions using the plugin headers provided)

Base build command:

```
| gcc -fplugin=./[plugin].so -O0 [reproducer].c -o base.bin
```

Variant build command:

```
| gcc -fplugin=./[plugin].so -O3 -flto [reproducer].c -o variant.bin
```

4.2 Reproducer Mapping

The following reproducer files (generated via yarpgen) can be used to confirm each finding:

Target Plugin	Reproducer File	Observed Symptom
stack_leak	reproducer_stack_leak.c	Verify disappearance of <code>is_alloca</code> and <code>large_stack_frame</code>
latent_entropy	reproducer_entropy.c	Audit <code>latent_entropy_execute</code> for correct inlining vs. deletion
struct_leak	reproducer_struct_leak.c	Check for zeroing instructions in the assembly at -O3

5. Recommended Remediation

The following actions are recommended in priority order. Each addresses a specific class of divergence identified in this audit:

Rec. 1 - Barrier Enforcement

Apply `__attribute__((noinline))` or volatile markers to core execution gates to prevent the optimiser from merging or eliminating security passes. This is particularly critical for `stackleak_cleanup_execute()` and `latent_entropy_execute()`.

```
| __attribute__((noinline)) void stackleak_cleanup_execute(void) { ... }
```

Rec. 2 - Dead Store Elimination Audit

Specifically audit `struct_leak` and `stack_leak` for cases where DSE may be removing security wipes perceived as useless writes. Consider annotating zeroing operations with a memory barrier or using `memset_s()` (which compilers are required to preserve).

Rec. 3 - Gate Visibility

Ensure gate functions (e.g., `stackleak_gate`) remain observable across compilation units. If the compiler cannot observe the gate from an external perspective, it is free to make static assumptions about which code paths deserve protection, potentially eliminating entire security checks.

PRIORITY: Recommendations 1 and 2 should be addressed before any production release targeting hardened kernel builds with `-O2` or higher.

6. Conclusion

This audit demonstrates that the Linux Kernel Hardening GCC plugin suite is systematically vulnerable to optimisation-induced divergence. The 'Likely Bug' verdict assigned to all six plugins reflects a pattern, not an isolated incident of security logic being silently erased or bypassed under standard production optimisation settings.

The absence of functional crashes makes this class of vulnerability particularly dangerous: existing test suites may pass while the security protections they validate are no longer present in the binary. Mandatory compiler barrier annotations and a targeted DSE audit are the minimum acceptable responses.